

MATHEMATICS

SECTION – A

Question numbers 1 to 6 carry 1 mark each.

1. Write the value of $\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$
2. If $A = \begin{pmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{pmatrix}$ and $BA = (b_{ij})$, find $b_{21} + b_{32}$.
3. Write the number of all possible matrices of order 2×3 with each entry 1 or 2.
4. Write the coordinates of the point which is the reflection of the point (α, β, γ) in the XZ-plane.
5. Find the position vector of the point which divides the join of points with position vectors $\vec{a} + 3\vec{b}$ and $\vec{a} - \vec{b}$ internally in the ratio 1 : 3.
6. If $|\vec{a}| = 4$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 6\sqrt{3}$, then find the value of $|\vec{a} \times \vec{b}|$.

SECTION – B

Question numbers 7 to 19 carry 4 marks each.

7. Solve for x : $\tan^{-1} \left(\frac{2-x}{2+x} \right) = \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right)$, $x > 0$.

OR

Prove that $2 \sin^{-1} \left(\frac{3}{5} \right) - \tan^{-1} \left(\frac{17}{31} \right) = \frac{\pi}{4}$.

8. On her birthday Seema decided to donate some money to children of an orphanage home. If there were 8 children less, every one would have got 10rs more. However, if there were 16 children more, every one would have got 10rs less. Using matrix method, find the number of children and the amount distributed by Seema. What values are reflected by Seema's decision?
9. If $x = e^{\cos 2t}$ and $y = e^{\sin 2t}$, prove that $\frac{dy}{dx} = \frac{y \log x}{x \log y}$.

OR

Verify Mean Value theorem for the function $f(x) = 2 \sin x + \sin 2x$ on $[0, \pi]$.

10. Show that the function f given by $f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & \text{if } x \neq 0 \\ -1, & \text{if } x = 0 \end{cases}$ is discontinuous at $x = 0$.

11. Find the equation of the tangent line to the curve $y = \sqrt{5x - 3} - 5$, which is parallel to the line $4x - 2y + 5 = 0$.

12. Evaluate : $\int_1^5 (|x - 1| + |x - 2| + |x - 3|) dx$

OR

Evaluate : $\int_0^\pi \frac{x \sin x}{1 + 3 \cos^2 x} dx$

13. Find : $\int \frac{2x+1}{(x^2+1)(x^2+4)} dx$